



Yanny or LoRa CANADA



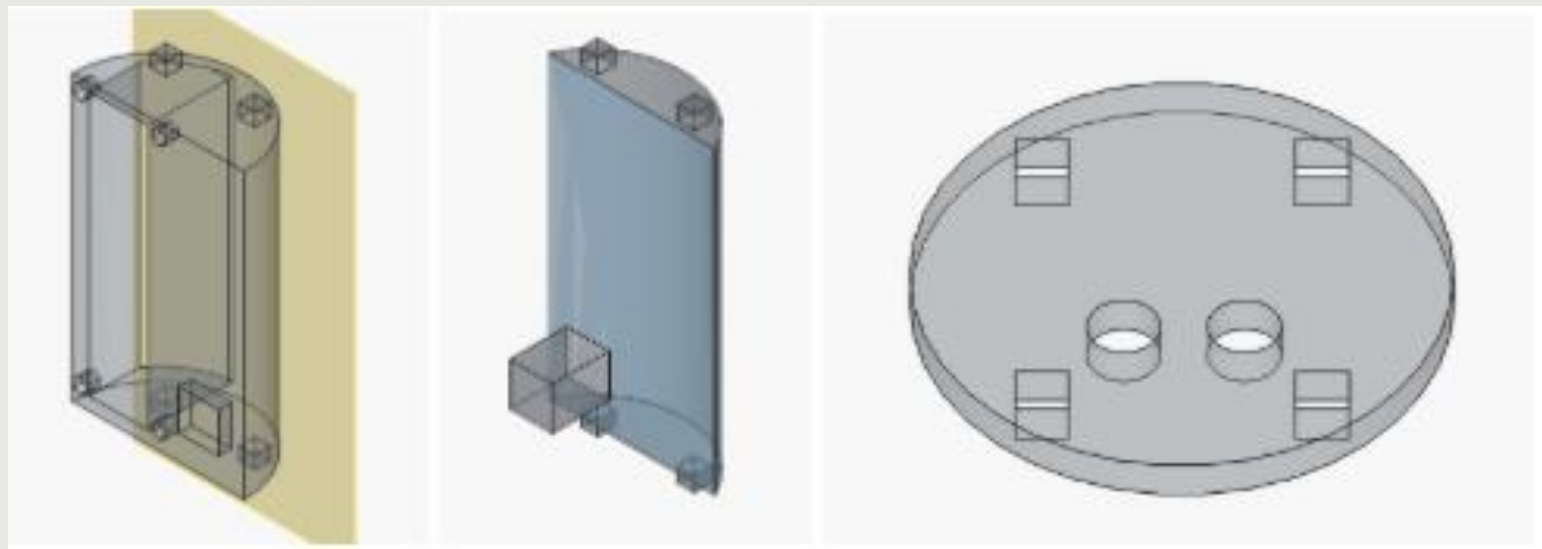
PRIMARY MISSION

To measure temperature and pressure once per second

SECONDARY MISSION

To create an SOS radio communication system that includes a hand generator, using the CanSat to extend the range to several kilometers.

DESIGN AND COMPONENTS



Our CanSat was designed with 3 devices in mind: the ground station, SOS radio device, and the CanSat itself. The LilyGO T-Beam V1.2 ESP32 was chosen as our microcontroller for all 3 devices, for environmental data collection and long-range communication relay.

To build our frames, we used PLA to maintain a lightweight design for our CanSat, while our SOS radio device used nylon as it better protects against external damages. Other components we used includes an OLED display, BME280, and heartbeat sensor on the SOS device to check the user's health before sending the data to ground station. For our landing system, we used a circular parachute using nylon fabric to prevent ripping during the descent and ensure ease during deployment. Using metal bars and a motor, we also designed a hand crank generator that pairs with our SOS device, ensuring access to the device in remote locations.

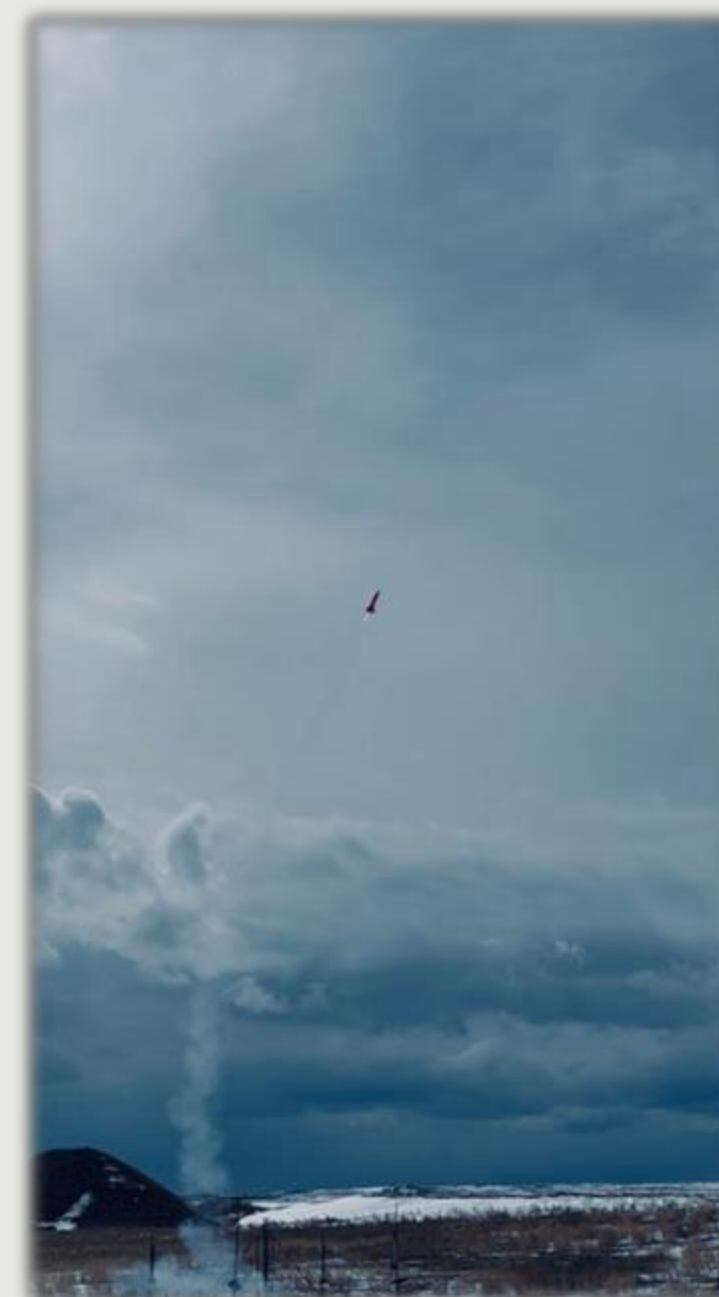


RESULTS

Time	Temp (C)	Humidity (%)	Pressure (hPa)	Altitude (m)	Lat	Lon
15:44:23	10.9	83.6	906.8	3.1	49.445186	-112.010757
15:44:35	10.7	83.6	866.3	386.8	49.446325	-112.013178
15:44:36	10.6	83.6	864.4	404.7	49.44681	-112.013385
15:44:38	10.2	83.5	861.6	431.7	49.447211	-112.013556
15:45:11	2.8	82.7	866.2	387.3	49.448768	-112.013149
15:45:23	1.3	82.6	873.2	320.1	49.448592	-112.013247
15:45:24	1.4	82.6	874.1	311.2	49.448575	-112.013229
15:45:26	1.6	82.6	875.9	294.6	49.448533	-112.013198
15:45:52	0.6	82.5	891.8	143.3	49.448109	-112.013965
15:46:00	1.5	82.6	897.7	85.4	49.447947	-112.014187
15:46:01	0.9	82.5	897.8	87.2	49.447929	-112.014265
15:46:02	0.4	82.5	897.6	88.7	49.44794	-112.014324
15:46:04	0.3	82.5	898.5	80.4	49.447906	-112.014377

Our launch conditions were not ideal, with strong winds, a thick cloud cover, and snow. Nonetheless, we successfully received 12 data packets through our LoRa radio system. Halfway through the launch, we lost communication for about 40 seconds and was able to reconnect when it was descending at an altitude of 387.3 metres.

The flight duration was around 150 seconds, the highest recorded altitude reached was 431.7 metres, and the largest distance recorded between the ground station and the CanSat was 599.7m. The descent rate calculated was estimated to be around 5.8m/s. All temperature and pressure data from our BME280 sensor was received successfully in the 12 data packets.

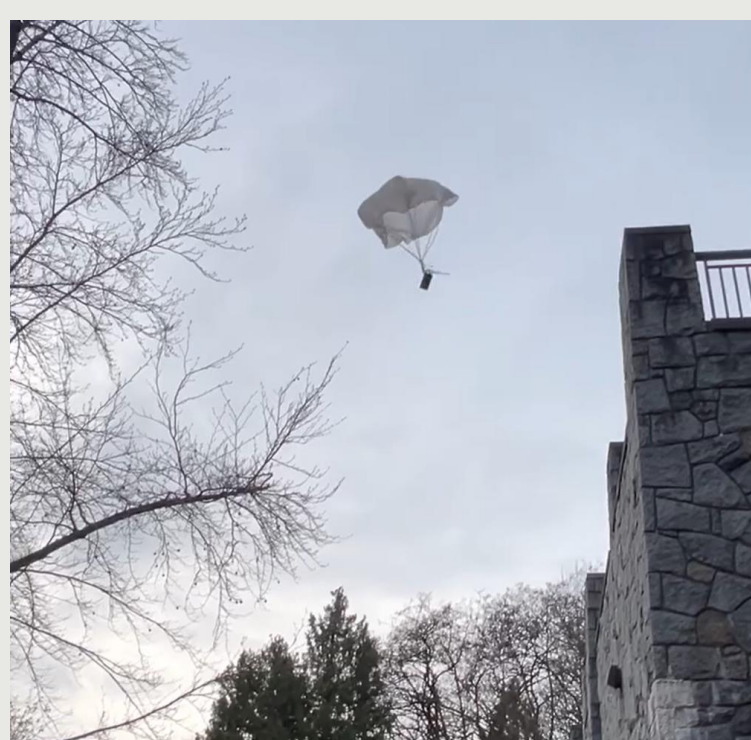


Our secondary mission was very successful. We received 7 packets of data from the SOS radio device, retrieving data on GPS location, temperature, and heart rate.

After retrieving our CanSat, we noticed water drops from melted snow inside. Fortunately, everything was still functional, and our CanSat did not break.



TESTING



We tested our CanSat extensively, and it was a significant part of our design process.

Firstly, we tested the hardware, including the BME280 sensor, heartbeat clicker, and GPS module. To test, we first had to solder header pins to the LoRa32 board. For some members of our group, it was our first time using Arduino and soldering.

Next, we tested the LoRa communication range by walking in different directions. After confirming a range of around 500–700 meters in dense urban regions with hills, we transferred the BME280 data between the two LoRa32s.

Finally, we tested our parachute landing system using failed frames with weights to find its descent velocity. We walked up a hill and dropped the parachute from 20 meters high. We divided this height by the average time taken for descent to estimate our terminal velocity. To ensure that the parachute would deploy properly under all conditions, we threw it in many different configurations at a variety of linear and rotational speeds.



LESSONS LEARNED

- The tolerance for PLA has to be a minimum of .25mm and .200 for a perfect fit.
- Designing a PCB would have been a lot for efficient space-wise, removing unnecessary wiring.
- Functionality should be prioritized over aesthetics.

